

JeevanMitra 2.0

A Verification-First, Multi-Layer Generative AI System for PM-AJAY Livelihood Matching & District Planning

ORGANIZATION	DEPARTMENT	COMPONENT	THEME
Ministry of Social Justice & Empowerment	Dept. of Social Justice & Empowerment	Grant-in-Aid (GIA), PM-AJAY	Agriculture, FoodTech & Rural Development

What changed in this version. This revision repositions the solution around two claims a generic voice-assistant pitch cannot make: (1) a hard guarantee that the system never implies training availability it hasn't verified, and (2) a planning-intelligence layer that turns every beneficiary conversation into GIA supply-gap evidence for district officials. The single AI/LLM block from the earlier draft is broken into six accountable generative layers, each with its own job, guardrails and failure mode, and it is explicitly positioned against the existing Skill India Assistant (SIA) rather than pretending to be the first of its kind.

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1. Problem Statement (as issued)

Problem Statement ID 26097. AI-driven voice assistant for livelihood mapping and NSQF-aligned skilling recommendations for SC communities under the GIA component of PM-AJAY.

PM-AJAY funds skilling and livelihood projects for Scheduled Caste households under its Grant-in-Aid component. Low digital literacy, limited awareness of modern trades, language barriers and text-heavy processes routinely mismatch beneficiaries with training that doesn't fit their interests, mobility or the local job market — producing high dropout, weak placement outcomes, and little real data for districts to plan with.

Basic issues named in the problem statement

Issue	What it looks like on the ground
Planning	Weak roadmap from perspective planning to execution; Annual Action Plans rarely grounded in real demand data.
Human resources	Difficulty identifying eligible participants and qualified trainers / financial counsellors.
Placement	Little linkage between finishing training and actually getting a job or starting an enterprise.
Coordination	Corporations, ministries and departments run separate processes with limited shared data.
Ground support	Inadequate technical and support staff at block and village level to guide beneficiaries.

Expected solution (as issued): an AI-powered multilingual voice assistant that helps SC beneficiaries identify suitable skill training and livelihood opportunities through voice conversation in regional languages and dialects, instead of text-based forms.

2. Why This Is Hard — And Why Most Solutions to It Fail Quietly

Most teams that take on this problem statement build a competent voice chatbot and stop. That fails in the field for two reasons this proposal treats as first-class design constraints rather than footnotes:

- **The trust problem.** A chatbot that recommends a qualification isn't the same as one that confirms a beneficiary can actually get into a batch near them. Confusing the two is exactly how beneficiaries end up travelling 40 km for a course that doesn't exist — the same failure mode the problem statement blames for high dropout.
- **The planning problem.** Even a perfect chatbot only fixes the beneficiary's experience. The problem statement's five "basic issues" are institutional (planning, coordination, ground staff) — a beneficiary-only tool does nothing for the Ministry unless the conversations are turned into district-level evidence.

JeevanMitra 2.0 is designed around closing both gaps explicitly, using generative AI as the connective tissue between conversation, verification and planning — not just as a chat interface.

3. The Two Claims This Solution Actually Makes

Claim 1 — The Verified Match Protocol

The system enforces a hard state machine on every recommendation it shows. A recommendation can only ever be in one of two visible states, and the beneficiary always sees which one:

State	Meaning	What the beneficiary sees
Interest Match	Qualification fits the beneficiary's profile per NQR data; local availability not yet confirmed by a field worker.	"Qualification match — local batch not confirmed." No apply button.
Verified Match	Same qualification AND a field worker has confirmed a live local batch, seat, or employer opening within the stated date window.	"Apply now" / "Request enrolment" becomes active.

No generative layer in this system is permitted to upgrade an Interest Match to a Verified Match. That transition can only be written by a human field worker action, logged in `audit_events`. This is enforced at the data layer, not just the prompt layer, so a prompt-injection or hallucination cannot silently promise a seat that doesn't exist.

Claim 2 — The Planning Loop

Every beneficiary conversation is simultaneously a service interaction and a data point. Aggregated, anonymised demand signals feed a generative reporting layer that produces a plain-language, block-level supply-gap brief for district officials automatically, on a set cadence — turning conversations that would otherwise sit in a database into something a planner would actually read before drafting an Annual Action Plan.

4. JeevanMitra vs. Skill India Assistant (SIA)

SIA (MSDE) already offers a multilingual, WhatsApp-based conversational assistant for course discovery and job listings. JeevanMitra is not pitched as a replacement for SIA or a reinvention of voice-based skilling discovery — the comparison below is included specifically to make clear where the two do not overlap.

	Skill India Assistant	JeevanMitra 2.0
Audience	Any citizen, any scheme	SC beneficiaries under PM-AJAY GIA specifically
Primary output	Course / job discovery answers	A tracked case file: profile → referral → enrolment → outcome
Availability handling	Not availability-checked	Hard-enforced Interest Match / Verified Match split
Institutional loop	None back to planners	District supply-gap briefs feed Annual Action Plans directly
Human role	None built in	Mandatory field-worker sign-off before any referral counts

Where useful, JeevanMitra's interview and matching backend is designed to sit behind the same WhatsApp/IVR channels SIA already validated at scale, rather than re-proving that voice-based, low-bandwidth delivery works in India — that groundwork is treated as settled, not as this project's contribution.

5. Solution Architecture — Six Generative AI Layers

The earlier draft described "an LLM" doing most of the work in one block. This version separates generative AI into six layers with distinct jobs, distinct guardrails, and distinct failure behaviour. Each layer is named so it can be independently tested, demoed and, if needed, replaced.

Layer 1 Conversational Intake Layer

Runs the voice interview itself: ASR for Hindi and regional dialects (incl. code-mixed speech), one-question-at-a-time dialogue management, and empathetic TTS for playback.

Generative task: Turning a single structured question into natural, dialect-appropriate spoken phrasing, and generating gentle re-prompts on unclear answers ("I heard you mention repair work — is that fixing machines, or something else?").

Guardrail: Intake never proposes qualifications, availability or advice — it only collects and confirms.

Layer 2 Structured Extraction Layer

Converts raw transcript into a structured beneficiary profile — education, prior skills, sector interest, mobility radius, work preference — as validated JSON, not free text.

Generative task: Entity extraction and intent classification from noisy, code-mixed transcripts; attaches a confidence score to every extracted field.

Guardrail: Any field below a confidence threshold is flagged for beneficiary confirmation before it is ever used in matching — the model cannot silently guess a low-confidence field into the profile.

Layer 3 Grounded Matching & Explanation Layer (RAG)

This is the layer that actually recommends. It runs deterministic hard filters plus a weighted score (interest 30% / prior skills 20% / feasible access 20% / verified local demand 20% / work preference 10%) against a retrieval index built only from the NQR qualification catalogue and the dated local-opportunity dataset — never from the model's own training knowledge.

Generative task: The LLM is only allowed to explain a ranked shortlist the retrieval system already produced, in the beneficiary's language, at a reading level appropriate to low literacy — it does not invent the shortlist itself.

Guardrail: Retrieval-augmented generation with a closed document set. If a qualification or opportunity isn't in the verified index, the model cannot mention it, by construction — this is the technical backbone of the Interest Match / Verified Match rule in Section 3.

Layer 4 Field-Worker Co-Pilot Layer

Assists the human reviewer, not the beneficiary. Drafts a plain-language case summary from the transcript, highlights low-confidence fields, and suggests (never applies) corrections.

Generative task: Summarising an 8-question interview into a 3-line case brief a field worker can scan in seconds, and drafting the outbound SMS/WhatsApp confirmation after a referral.

Guardrail: Every suggested correction and every referral action requires an explicit worker click; nothing this layer drafts is sent or saved without human approval.

Layer 5 Planning Narrative Layer

Generates the district-level supply-gap brief described in Claim 2 — turning aggregated, anonymised demand-vs-capacity tables into a short written brief a district officer can read before a planning meeting.

Generative task: Example output: "Block Bahjoi: 180 SC youth requested agro-processing training this quarter; nearest sanctioned centre is 45 km away. Comparable gaps last quarter: none recorded. Suggested action: evaluate a mobile training unit." — narrative generated strictly from the aggregated numbers, with every figure traceable back to source rows.

Guardrail: Numbers in the narrative are inserted from the underlying query result, not generated by the model — the LLM writes the sentence, the database supplies every figure inside it.

Layer 6 Outcome & Drift Monitoring Layer

A lightweight layer that watches its own recommendation patterns over time for the bias and drift risks a scheme like this must guard against.

Generative task: Periodically drafts a short audit note flagging patterns worth human review — e.g. a gender skew in which trades are being recommended, or a spike in "local batch not confirmed" responses for one block.

Guardrail: Output is advisory only, routed to a human reviewer's queue; it never changes matching weights or beneficiary-facing content on its own.

6. End-to-End Workflow

Beneficiary-facing flow

Step	What happens
1. Welcome & consent	Language selection; short spoken explanation of what is collected, why, and how to stop. Consent flow aligned to DPDP Act notice-and-consent principles.
2. Voice interview (Layer 1)	One question at a time — district/block, education, current work, interests, prior skills, travel radius, accessibility needs, wage vs. self-employment preference. "Skip" and "I don't know" always allowed.
3. Structured profiling (Layer 2)	Transcript converted to a confidence-scored structured profile; low-confidence fields are read back for correction before anything else runs.
4. Confirm profile	Key answers read back aloud: "You said up to 10 km travel and self-employment — is that right?" Nothing proceeds without confirmation.
5. Grounded matching (Layer 3)	Hard filters + weighted scoring against the verified NQR/local-opportunity index. Output: up to three ranked options, each tagged Interest Match or Verified Match.
6. Explain & act	Each option explained in the beneficiary's language with the skill gap, qualification detail and honest availability status. Options to compare, or request a field-worker call.
7. Human review (Layer 4)	A field worker verifies uncertain facts, corrects the profile if needed, and is the only actor who can convert a referral into an enrolment record.
8. Outcome tracking	Enrolment, completion and employment/self-employment status logged against the case, visible to the beneficiary as a plain-language progress tracker.

Institutional / planning flow (runs in parallel, Layer 5 + 6)

- Every completed interview writes an anonymised demand record (trade interest, block, constraints) alongside the identifiable case record.
- A scheduled job aggregates demand records against sanctioned seats/capacity per trade per block.
- Layer 5 drafts a short narrative supply-gap brief per district, with every cited number traceable to the aggregation query.
- Layer 6 scans recent recommendation and outcome patterns for drift or bias signals and appends an advisory note.
- District officer reviews the brief inside the planning console, can drill into the underlying case-level data, and exports it for Annual Action Plan input.

7. Data Model & Backend Architecture

Next.js beneficiary app + field-worker dashboard + (later) IVR/WhatsApp adapter → API/orchestration layer (STT-TTS adapter, interview & extraction service, retrieval-grounded matching service, planning-narrative service, referral/reporting service) → Supabase Postgres + private file storage → verified NQR catalogue + maintained local opportunity data.

Core tables

Table	Essential fields
beneficiaries	id, preferred language, district/block, contact preference, created date
consents	beneficiary_id, purpose, notice version, choice, timestamp
interview_sessions	id, beneficiary, channel, status, last question, language
profile_answers	beneficiary_id, field, value, confidence score, confirmation status, source
qualifications	NQR identifier/link, title, sector, NSQF level, entry criteria, skills, verification date
local_opportunities	qualification ID, centre/employer, location, batch/vacancy status, source, verified date
recommendations	beneficiary, qualification, score, match_state (Interest/Verified), explanation, data snapshot
referrals	beneficiary, opportunity, assigned worker, status, next follow-up
outcomes	enrolment, completion, employment/self-employment status, follow-up date
planning_briefs	district, period, aggregation snapshot, generated narrative, reviewer sign-off
audit_events	actor, action, record, timestamp — including every match_state transition

Row-level permissions apply throughout: beneficiaries see only their own information, assigned workers see only their cases, district admins see appropriately scoped and aggregated records. Audio is stored privately, only when needed, under a defined retention policy. Caste documents and precise location are collected only where a defined eligibility-verification workflow requires them, and that collection is kept separate from the AI-facing profile tables.

8. Screen Flow Summary

Full wireframes are in the companion design document. In brief, the beneficiary journey runs: Welcome & Language → Consent → Voice Interview → Confirm Profile → Recommendations (tagged Interest Match / Verified Match) → Pathway Comparison → Training Centre Detail → Progress Tracker. The field-worker journey runs: Case Overview → Case Review & Correction → Approve Referral / Flag Ineligible. The district journey is a single Planning Console showing voice demand vs. sanctioned seats per trade, with the Layer 5 narrative brief attached.

Design principle carried through every screen: large controls, short spoken prompts, no mandatory typing, and a visibly different visual treatment for Interest Match vs. Verified Match cards so the distinction in Section 3 is impossible to miss, not just a backend flag.

9. Build Status, Roadmap & Demo Plan

Stated plainly, so scope is judged on what is real: mark each item honestly before presenting.

Component	Status	Notes
Voice interview (Layer 1+2), one language, sample question set	[fill in]	Target: complete without typing, end to end
Grounded matching against a small manual NQR sample (Layer 3)	[fill in]	Prove: recommendations change when interests/mobility change
Interest Match / Verified Match UI distinction	[fill in]	Core differentiator — prioritise this over extra channels
Field-worker review & referral dashboard (Layer 4)	[fill in]	Prove: a suggestion becomes a tracked, human-approved action
District planning brief (Layer 5)	[fill in]	Even a single generated sample brief is a strong demo beat
Drift/bias advisory notes (Layer 6)	Roadmap	Describe design; not expected in a hackathon build
IVR / WhatsApp channels	Roadmap	Explicitly Phase 2 — do not claim built

Suggested demo sequence

- Two beneficiaries, same district, different interests/mobility — show different ranked recommendations.
- One recommendation shown as Interest Match; a field worker confirms a batch live and it visibly flips to Verified Match.
- A worker corrects one misheard answer and the recommendation set updates immediately.
- The district console shows a real supply gap for one trade with no verified local batch, and the Layer 5 narrative brief generated from that exact data.

10. Governance, Privacy & Risk

Data, privacy & ethics

- Explicit, audio consent before data collection, aligned with DPDP Act notice-and-consent and correction/erasure principles.
- Data minimisation — only necessary fields collected; caste-eligibility documents handled in a separate, worker-only verification flow, not the AI profiling tables.
- Anonymised, aggregated data only for planning briefs (Layer 5); personally identifiable data restricted to authorised staff via row-level permissions.
- Bias mitigation — Layer 6 specifically screens for skewed recommendation patterns (e.g. by gender) and routes them to human review rather than auto-correcting silently.
- Accessibility — audio-only flows, slow-speech and repeat options for users with visual or other disabilities.

Risks & mitigations

Risk	Mitigation
Hallucinated availability or fabricated courses/jobs	Layer 3 is retrieval-grounded against a closed, verified document set; match_state upgrades require a logged human action.
Regional-dialect ASR accuracy	Confidence scoring on every extracted field (Layer 2); low-confidence fields are always confirmed with the beneficiary, never silently trusted.
Algorithmic or gender bias in recommendations	Explainable weighted scoring, reviewed by field workers; Layer 6 issues periodic drift/bias advisories for human audit.
Data privacy / DPDP compliance	Minimal collection, private audio storage with retention limits, role-scoped access, consent flows tied to notice versioning.
Technical failure in low-connectivity areas	IVR-first fallback design, kiosk queuing for later sync; no promise of fully offline cloud AI.
Over-claiming scope in front of evaluators	Section 9's build-status table is presented honestly, distinguishing built vs. roadmap items before any Q&A.

Disclaimer. All beneficiary names, case IDs, dashboard counters and figures used anywhere in this document and its companion design mockups are illustrative placeholders. A functional prototype built from this proposal is not an official PM-AJAY integration and does not guarantee admission, funding or employment.